

Examination Statistical Methods

Linköpings Universitet, IDA, Statistik

Course code and name:	732A83 Statistical Methods
Date:	2024/12/05, 8–12
Examinator:	Krzysztof Bartoszek
Allowed aids:	Pocket calculator Table with common formulæ and moment generating functions (distributed with the exam) Table of integrals (distributed with the exam) Tables with distributions, probabilities and percentage points (distributed with the exam) One double sided A4 page with own hand written notes
Grades:	A= $[18 - \infty)$ points B= $[16 - 18)$ points C= $[14 - 16)$ points D= $[12 - 14)$ points E= $[10 - 12)$ points F= $[0 - 10)$ points
Instructions:	Write clear and concise answers to the questions. Reporting on the exam a probability outside the $[0, 1]$ interval or a negative variance without a comment that a mistake must have been made somewhere will result in a NEGATIVE point for each such instance.

Problem 1 (3p)

For a certain laptop brand it is known that after 15 years of work 50% of them have problems with screens, 30% have problems with hard drives, and 20% have problems with both hard drives and screens.

1. (1p) Calculate the probability that a 15 year old laptop of this brand will have both a working screen and hard drive. Then, calculate the probability that a 15 year old laptop of this brand will have a broken hard drive if we know it has a broken screen.
2. (1p) Calculate the probability that a 15 year old laptop of this brand will have a working hard drive if we know it has a broken screen.

3. (1p) Is the event of a broken screen independent of the event of a broken hard drive? Justify your answer.

Problem 2 (2p)

The time to failure in years of a certain brand of RAM is a continuous random variable with density function

$$f(x) = \begin{cases} Cx^{-4} & x \geq 2, \\ 0 & x < 2. \end{cases}$$

1. (1p) Calculate the constant C and find the cumulative distribution function of the lifetime.
2. (1p) Find the expectation and variance of the time to failure.

Problem 3 (5p)

Let $X_1, \dots, X_n$ be independent and identically distributed from the following distribution with unknown parameter θ :

$$f_{\theta}(x) = \begin{cases} (\theta + 1)x^{\theta}, & 0 \leq x \leq 1, \\ 0, & x \notin [0, 1]. \end{cases}$$

1. (3p) Find the maximum likelihood estimator of θ .
2. (2p) Find the methods of moment estimator of θ .

Problem 4 (2p)

Assume that X_1, X_2, X_3 are random variables such that

$$\begin{aligned} \mathbb{E}[X_1] &= 3 & \text{Var}[X_1] &= 1 & \text{Cov}[X_1, X_2] &= 1 \\ \mathbb{E}[X_2] &= -5 & \text{Var}[X_1] &= 3 & \text{Cov}[X_1, X_3] &= -2 \\ \mathbb{E}[X_3] &= 0 & \text{Var}[X_1] &= 2 & \text{Cov}[X_2, X_3] &= 0 \end{aligned}$$

1. (1p) Find $\mathbb{E}[3X_1 + 2X_2 + 10^8 X_3]$.
2. (1p) Find $\text{Var}[2X_1 + X_2 + 4X_3]$.

Problem 5 (3p)

The number of connections per minute to a certain server follows a Poisson distribution with mean λ . The server's administrator decides to use Bayesian statistics to make inference on λ , using the gamma distribution with parameters a and b as a prior, i.e., $\lambda \sim \Gamma(a, b)$.

1. (2p) Find the posterior distribution of λ , and an expression for the Bayesian estimator, $\hat{\lambda}$.
2. (1p) Suppose that the server's administrator collected data for 5 minutes and observed a total of 300 connections. What would be the estimate for the average number of connections per minute to the server, if the $\Gamma(10, 2)$ distribution is used as a prior?

Problem 6 (3p)

A Markov chain has the following transition probability matrix

$$P = \begin{bmatrix} 0.2 & ?? & 0.3 \\ ?? & 1 & ?? \\ 0 & 0 & ?? \end{bmatrix}$$

1. (1p) Fill in the missing values, i.e., those marked with ??.
2. (1p) Find the stationary distributions of this Markov Chain.
3. (1p) Does this chain have an absorbing state?

Problem 7 (2p)

It is known that malicious connections to a server follow a Poisson process with a rate of 4 attacks per minute.

1. (1p) Find the probability that in 2 consecutive minutes no attacks took place.
2. (1p) Find the expectation and variance of the time between two attacks.